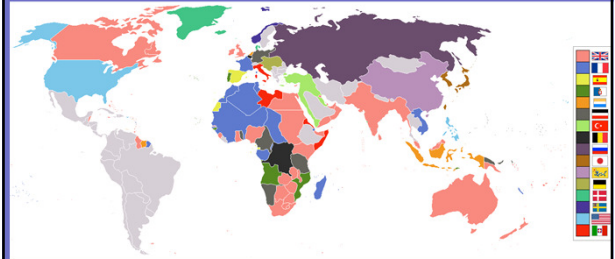


Skin infections in travellers

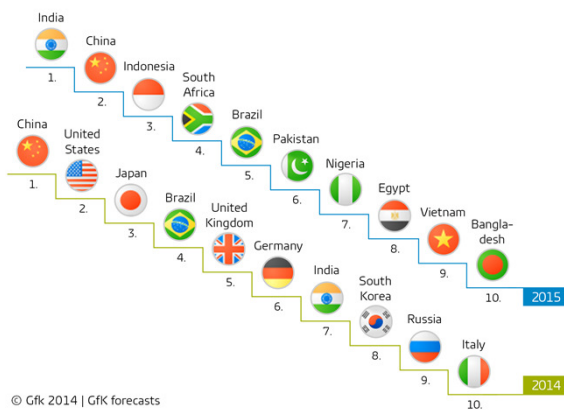
Henry J.C. de Vries
Dermatology AMC, University of
Amsterdam
STI outpatient clinic, Health service
Amsterdam

Political map in 1914



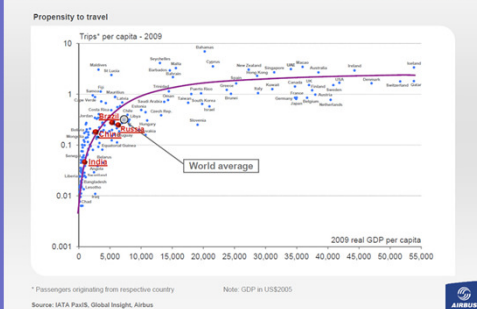
2

Top 10 smartphone markets for growth by value, 2015 compared with 2014



Travel and globalization

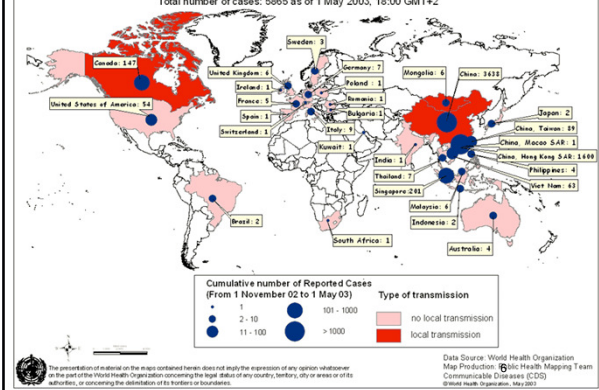
Emerging economies on the edge to strong travel growth



Refugee crisis in Europe



SARS : Cumulative Number of Reported Probable Cases



GeoSentinel Surveillance of Illness in Returned Travelers, 2007–2011

Table 1. Characteristics of Ill Returned Travelers

Variable	Total	Gastrointestinal Diagnoses	Febrile Illness	Dermatologic Diagnoses	Respiratory or Pharyngeal Diagnoses	Neurologic Diagnoses	GU, STI, and Gynecologic Diagnoses
Travelers, n (%)	42 173	14 346 (34.0)	9817 (23.3)	8227 (19.5)	4613 (10.9)	724 (1.7)	1209 (2.9)
Diagnoses, n*	49 379	14 837	10 092	9669	4851	738	1260
Men, %†	49.9	44.5	58.9	47.8	51.7	50.6	37.2
Median age (range), y‡	34 (0–95)	32 (0–92)	35 (0–91)	35 (0–95)	36 (0–93)	38 (0–88)	37 (0–88)
Travel reason, %§							
Tourism	55.7	59.3	45.1	68.2	53.6	55.4	51.9
Business	13.6	14.2	14.2	9.5	17.0	12.8	13.9
Visiting friends/relatives	15.5	8.8	28.1	10.2	16.5	13.5	18.5
Missionary	11.6	13.9	8.7	8.4	9.2	13.5	13.1
Student	2.6	3.4	2.0	2.5	2.9	4.0	1.9
Region, %							
Australia and New Zealand	0.5	0.2	0.2	0.7	1.8	0.7	0.6
Southeast Asia	16.3	13.8	18.1	22.0	17.4	10.1	17.3
South-Central Asia	13.6	19.1	13.2	9.1	10.6	7.6	11.1
Northeast Asia	2.7	2.2	1.2	2.9	5.8	3.2	2.4
Europe	4.7	3.5	2.1	4.7	10.1	9.3	7.4
Latin America and Caribbean	19.2	20.4	14.3	27.3	14.2	23.6	15.6
Middle East and North Africa	6.1	8.7	2.5	5.6	5.2	6.5	6.1
North America	1.5	0.5	0.4	1.6	5.2	2.9	2.1
Oceania	0.8	0.7	1.0	1.2	0.9	1.4	0.5
Sub-Saharan Africa	26.7	22.5	42.6	19.5	20.6	22.3	26.9

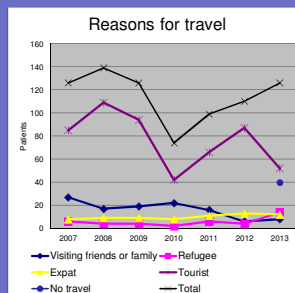
Table 3. Notable Specific Diagnoses Among Ill Returned Travelers

Diagnoses	Cases, n	Median Age, y	Man–Woman Ratio	Reason for Travel, %*			Reported Pretravel Visit, %	Top Countries of Exposure†
				Tourism	Visiting Friends/Relatives	Business		
Dermatologic								
Rabies PEP after bite or scratch	1249	31	1.0	68.9	17.7	7.1	25.5	Thailand, Indonesia, China, and India
Cutaneous larva migrans	806	30	0.9	80.8	2.9	5.0	42.1	Thailand, Brazil, Mexico, and Malaysia
Leishmaniasis (cutaneous or mucocutaneous)	264	23	1.9	49.6	17.0	7.2	49.6	Bolivia, Afghanistan, and Costa Rica
Myiasis	174	36	1.3	71.8	6.3	11.5	46.9	Senegal, Brazil, Costa Rica, and Belize
Tungiasis	87	30	1.1	52.9	5.7	10.3	54.0	Brazil, Madagascar, Uganda, and Ethiopia
Gnathostomiasis	12	26.1	1	50.0	8.3	8.3	33.3	Cambodia and Indonesia (92% from Asia)
Leprosy	11	44	4.5	90.9	–	9.1	18.2	Pakistan and Vietnam
Cutaneous atypical mycobacteria	6	36.5	2.0	50.0	33.3	16.7	16.7	All different (Caribbean, sub-Saharan Africa, and South-Central Asia)
Sporotrichosis	1	14	1 woman	–	100	–	0	Mexico
Yaws	1	67	1 woman	–	100	–	0	Jamaica

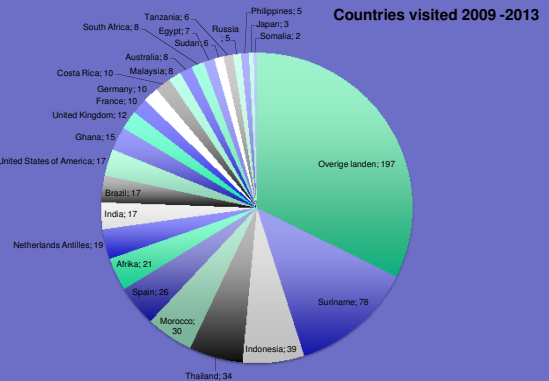
8

Outpatient clinic for skin infections AMC Patient population

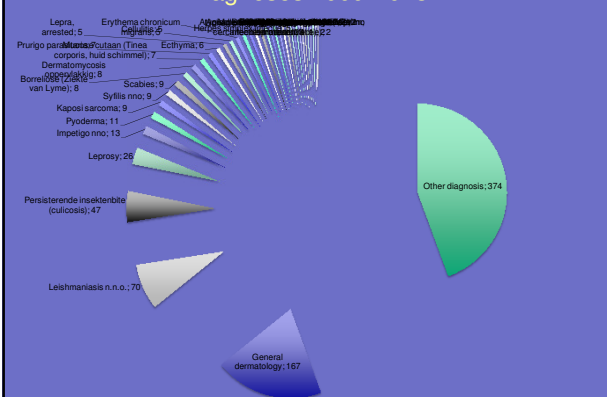
- Mondays
- Skin disorders in travellers
- Training of residents
- Referral centre
 - dermatologists
 - Amsterdam GP's
 - Outpatient clinic tropical medicine AMC
 - 020-5662387 (line for professionals)
 - 020-5662530 (line for patients)
- Ex leprosy patients (app. 70)
 - Neuropathy
 - Multidisciplinary team rehabilitation medicine and surgery



AMC dermatology outpatient clinic for skin infections Countries visited 2009–2013



Outpatient clinic for skin infections Diagnoses 2009–2013



Overview

- Skin anatomy and function
- Skin types and skin diseases
- Climate and skin
- Exanthema
- Ulcers
- Itch

Overview

- Skin anatomy and function
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Skin functions 1

- Barrier:
 - Fluid and electrolyte loss
 - UV light
 - Microbes
 - Mechanical barrier
 - Natural microflora
 - Innate immunity
 - Adaptive immunity
 - Mechanical stress



Skin functions 2

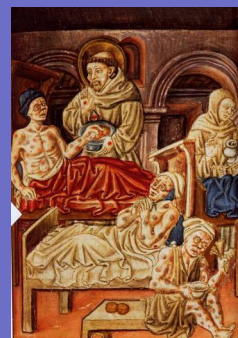
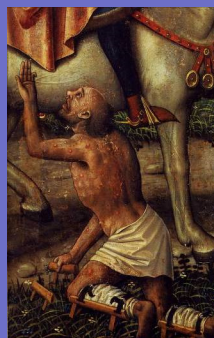
- Thermostasis:
 - Releasing energy(sweating, vasodilatation)
 - Preserving energy (vasoconstriction)
 - Generating energy (shivering)
- Vitamin D production

Skin functions 3

- Sensation
 - touch
 - pain
 - temperature
 - itch
- Communication
 - Attraction
 - Selection



Skin and stigma



Skin as 3 dimensional organ

- Naked eye:
– 2 dimensional info
- Skin biopsy:
– 3 dimensional info



Skin as 4 dimensional organ

- “life of lesions”



WHO.
I. Arita.

Life of lesions



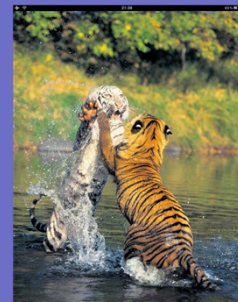
Overview

- Skin anatomy and function
- **Skin types and skin diseases**
- Climate and skin
- Exanthema
- Ulcers
- Itch



Skin types

- Genes
 - Feomelanin
 - Eumelanin
- Circulation
- Diet



THE FITZPATRICK SKIN TYPE SCALE					
I IVORY	II BEIGE	III LIGHT BROWN	IV MEDIUM BROWN	V DARK BROWN	VI VERY DARK BROWN
Characteristics Pale skin, light or red hair, prone to freckles. Burns very easily and rarely tans.	Characteristics Fair skin, likely to have light hair, blue or brown eyes. Some have dark hair but still have a fair skin. Usually burns but may gradually tan.	Characteristics Light olive skin with dark hair and brown or green eyes. Burns with long exposure to the sun but generally tans quite easily.	Characteristics Brown eyes and dark hair. Burns with very lengthy exposures but always tans easily.	Characteristics Naturally brown skin, brown eyes and dark hair. Burns only with excessive exposure to the sun. Skin easily darkens further.	Characteristics Black skin with dark brown eyes and black hair. Burns only with extreme exposure to the sun. Skin very easily darkens further.
Sun protection At the greatest risk of developing skin cancer. Needs to protect skin, preferably with clothing.	Sun protection At the greatest risk of developing skin cancer. Needs to protect skin, preferably with clothing.	Sun protection Should protect themselves in strong sunshine.	Sun protection Should protect themselves in strong sunshine.	Sun protection Protect themselves when outdoors in the sun for a long time.	Sun protection Protect themselves when outdoors in the sun for a long time.

The skin color chart of Felix von Luscha

1	10	19	28
2	11	20	29
3	12	21	30
4	13	22	31
5	14	23	32
6	15	24	33
7	16	25	34
8	17	26	35
9	18	27	36

Skin types classification

Skin Type	Sunburn Tendency	Under sun exposure	Skin, Hair, and Eye Color
I	I always get a sunburn.	I never get a tan.	white skin, freckles, blond or red hair, blue or green eyes
II	I usually get a sunburn.	I sometimes get a tan.	white skin, blond hair, blue or green eyes
III	I seldom get a sunburn.	I usually get a tan.	white skin, usually dark hair, and brown eyes
IV-VI	I never get a sunburn.	I always get a dark tan.	brown to dark skin/ brown or black hair/ brown eyes

Allergic eczema



Naevus flammeus



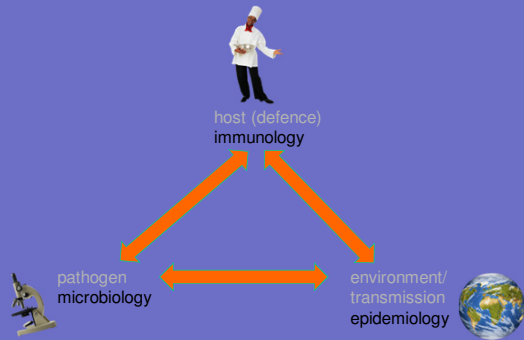
Contact allergy



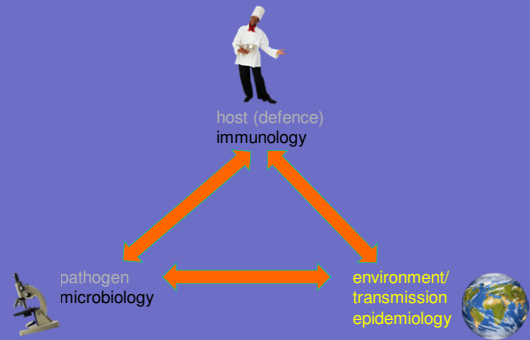
Overview

- Skin anatomy and function
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- Ulcers
- Itch

Driving factors of infection



Driving factors of infection



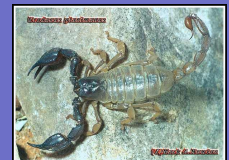
Common skin pathology in arid climates

- Fotodermatoses
- Ecthyma (desert sore)
- Cutaneous diphteria
- Bristles, spikes, prickles (Sabra dermatitis)
- Urticaria multiformis
- endemica (Harara)
- Frost bite



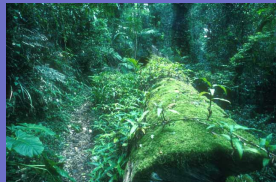
Common skin pathology in semi arid climates

- fotodermatoses
- insect bites
- scorpion and spider bites
- actinomycotic mycetoma



Common skin pathology in humid climates

- Fungal infections
- Bacterial infections
- Parasital infections
- Insect bites



Human interference



Extremely dry conditions

- Itch
- Fissures
- Emollientia
 - Vaseline/paraffin
 - Cetomacrogol ointment
- Shower limitation
 - Less often
 - Less long
 - Less hot
 - Less soap



Skin conditions due to heat

- Miliaria
 - Sweat gland inflammation
 - Miliaria rubra (prickly heat)
 - Miliaria pustulosa
 - Miliaria profunda
- Urticaria

Miliaria rubra



Miliaria profunda



Miliaria pustulosa



folliculitis



miliaria pustulosa



Heat induced urticaria



UV induced skin conditions

- Sun burn
- Polymorphic light eruption
 - (misnomer “sun allergy”)
- Phototoxic eruption
- Photoallergic eruption

Sun burn



Polymorphic light eruption

- First UV exposure
 - Spring
 - High altitude (skiing, hiking)
- Extending body parts (ears, nose)
- Folds (elbow, neck)
- Diminishes with repeated exposition
- Polymorphic (many forms)
 - Papules, spots, vesicles, excoriations)
- Gradual UV exposure
 - Sunprotection

Polymorphic light eruption



Phototoxic reaction



Common photo sensitizers

tetracyclines (high-low)	quinolones (high-medium)	NSAID's
demeclocycline: high	ciprofloxacin	carprofen
doxycycline: medium	levofloxacin	ibuprofen
tetracycline: low	norfloxacin	naproxen
	ofloxacin	proxicam
		tiaprofenic acid

Phyto-phototoxic dermatitis



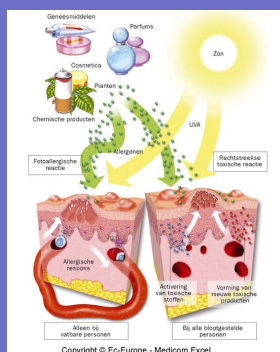
Local contact with furocoumarines: plants - perfumes

Fotoallergic reaction

- sulfonamides
- antiseptics
- promethazine
- chlorthiazide
- chlorpromazine
- chlorpropamide
- tolbutamide



Fototoxicity vs fotoallergy



dd photosensitivity

	phototoxicity	photoallergy
pathophysiology	direct tissue injury	type 4 allergy (DTH)
clinical picture	sunburn	eczema
onset	min-hrs	days
dosage	dependent	independent
sensibilization	no	yes
cross reactivity	rare	common
susceptibility	everyone exposed	only susceptible cases

Break



Overview

- Skin anatomy and function
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Fever and rash

- Rickettsial fevers
- Dengue fever
- Typhoid fevers

Rickettsiosis

- Travellers
- Intracellular bacterial infection
- Endothelium
- Vasculitis
- Vector borne primarily ticks
- Less common: louse, fleas and mites

Rickettsiosis

Antigenic Group	Disease	Species	Vector	Animal Reservoir(s)	Geographic Distribution
Anaplasma	Human granulocytic anaplasmosis	<i>Anaplasma phagocytophilum</i>	Tick	Deer, elk, small mammals, and rodents	Worldwide
Ehrlichia	Human monocytic ehrlichiosis	<i>Ehrlichia chaffeensis</i>	Tick	Deer, wild and domestic dogs, domestic ruminants, and rodents	Worldwide
Ehrlichiosis		<i>Ehrlichia ewingii</i>	Tick	Deer, wild and domestic dogs, and rodents	North America
Scrub typhus	Scrub typhus	<i>Orientia tsutsugamushi</i>	Larval mite (chigger)	Rodents	Asia-Pacific region from Maritime Russia and China to Indonesia and North Australia to Afghanistan
Spotted fever	Rickettsiosis	<i>Rickettsia aeschlimannii</i>	Tick	Unknown	South Africa, Morocco, Mediterranean littoral
	African tick bite fever	<i>Rickettsia africae</i>	Tick	Rodents	Sub-Saharan Africa, West Indies
	Mediterranean spotted fever or Boutonneuse fever	<i>Rickettsia conorii</i>	Tick	Dogs, rodents	Southern Europe, southern and western Asia, Africa, India
	Cat flea rickettsiosis	<i>Rickettsia felis</i>	Cat and dog flea	Domestic cats, rodents, opossums	Europe, North and South America, Africa, Asia
	Rocky Mountain spotted fever	<i>Rickettsia rickettsii</i>	Tick	Rodents	North, Central, and South America
Typhus fevers	Epidemic typhus, Sympatric typhus	<i>Rickettsia prowazekii</i>	Human body louse	Humans, flying squirrels	Central Africa, Asia, Central, North, and South America
	Murine typhus	<i>Rickettsia typhi</i>	Flea	Rodents	Tropical and subtropical areas worldwide

Rickettsiosis symptoms

- within 1–2 weeks of exposure
 - fever, headache, malaise, sometimes nausea and vomiting
- maculopapular, vesicular or petechial rash
- eschar at the site of the tick bite
- mild or moderate illness
- except epidemic typhus and RMSF
 - may be fatal in 20%–60% of untreated cases
- require further tests to make an accurate diagnosis.

Rickettsiosis clinical presentation



Diagnosis based on serology

PCR skin biopsy eschar (specialised labs)

Immune fluorescence and ELISA:
low predictive value
non species specific

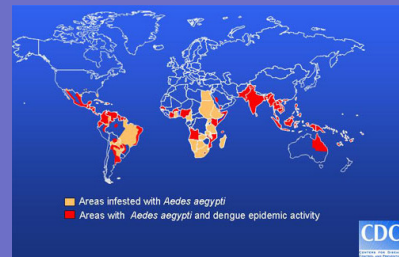
Western Blot and specialized ELISA (specialised labs)

Does it influence your clinical management?
Tx: doxycycline

Dengue fever

- 1,2 million cases reported
- Estimated 51 million cases
- Group B arbovirus
- Transmitted by mosquitos: genus aedes
- Macular/maculopapular/hemorrhagic

Dengue, geographical distribution



Dengue fever

- Dengue fever
- Dengue hemorrhagic fever
- Dengue shock syndrome



Eurosurveillance
Europe's journal on infectious disease epidemiology, prevention and control

HOME ARCHIVES ABOUT US EDITORIAL POLICY FOR AUTHORS LINKS

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Advanced search

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Eurosurveillance, Volume 15, Issue 39, 30 September 2010

Rapid communications

FIRST TWO AUTOCHTHONOUS DENGUE VIRUS INFECTIONS IN METROPOLITAN FRANCE, SEPTEMBER 2010

G La Roche (g.laruche@invs.sante.fr)¹, Y Souarès¹, A Armengaud², F Peloux-Petit³, P Delaunay⁴, P Desprès⁵, A Lenglet⁶, F Jourdain¹, I Leparo-Goffart⁶, F Charlet⁶, L Ollier⁴, K Mantey², T Mollet², J P Fournier⁴, R Torrents², K Lellimeyer², P Hilaire⁶, H Zeller⁶, W Van Bortel⁶, D Dejourn-Salamanca¹, M Grandadam⁶, M Gastellu-Etchegorry⁷

• Nice, south-east France,
• *Aedes albopictus*
• Evidence of dengue virus circulation in this area.

Other tickborne infections

- Borreliosis
- Babesiosis
- Tularemia
- Ehrlichiosis
- Anaplasmosis (skin sympt rare)
- Pogosta disease (virus)

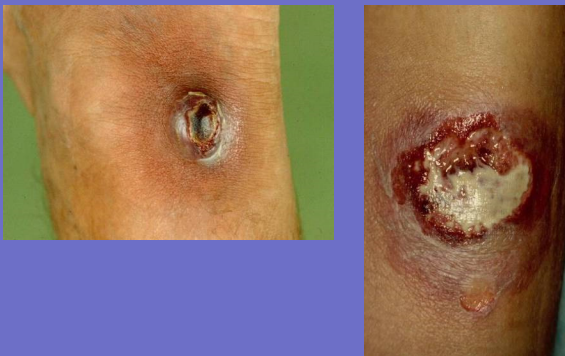
Overview

- Skin anatomy and function
- Skin types and skin diseases
- Climate and skin
- Exanthema
- **Ulcers**
- Itch

Ulcers:

- streptococcal ulcer
- ecthyma
- endemic treponematoses
- cutaneous diphtheria
- toe web infection (trench foot)
- cutaneous anthrax
- leishmaniasis

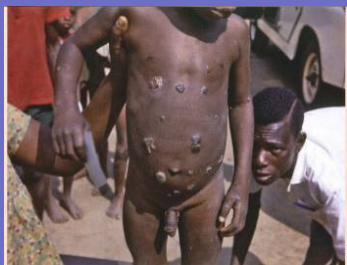
Streptococcal ulcer



Ecthyma



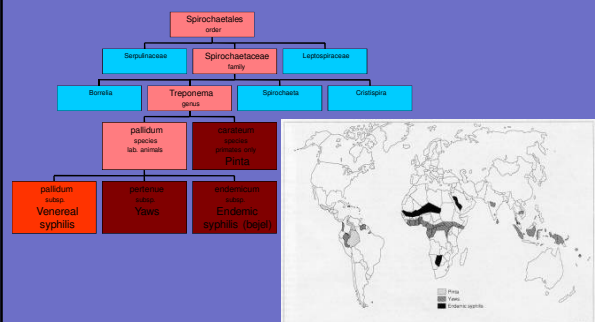
Endemic treponematoses/ non venereal syphilis



Source: Medisch Contact 2007

77

Epidemiology endemic treponematoses/ non venereal syphilis



78

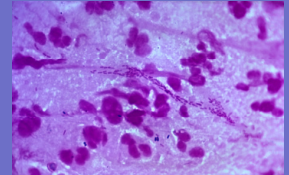
22 y.o. male student

- Minor abrasion during jungle trip in Ambon, Indonesia
- Ulceration 5 weeks later
- No improvement after flucloxacillin 10 days



H. ducreyi cutaneous ulcer

- Known as STI agent for genital ulcers
- In yaws-endemic communities
 - South-East Asia and the Pacific



Cutaneous diphtheria



Corynebacterium diphtheriae

Skin and Soft Tissue Infections in Immigrants (Patera Foot)

- Previously healthy young immigrants from sub-Saharan Africa
- After crossing the sea on small boats (*patera*).
- Suggestive for infectious cause organism yet to be determined



A) Initial severe skin and soft tissue infection of the lower limbs;
B) extensive debridement of the left foot
C) left foot after skin allograft.

Hugo-Guillermo Tornavasio-de la Vega Emerg Inf Dis 2009

Toe web mixed infection (trench foot)

- First described in WW I in military in trenches
- Maceration
 - Humidity
 - Long marches
 - Poor hygiene
- Mycosis interdigital spaces
- Secondary infection Gram negative organisms
 - *P. aeruginosa*
- Extensive maceration



Toe web mixed infection (trench foot)

- Hygienic measures
 - Keep dry
 - Cotton fabric
 - Remove necrosis and hyperkeratosis
- Local treatment
 - Antimycotic ointment + Zilver sulfadiazine
- Extensive treatment
 - Ciprofloxacin bid 500 mg 7-14 days + antimycotic ointment
 - (Highly potent corticosteroid ointment)



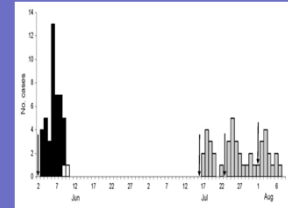
Cutaneous anthrax

- *Bacillus anthracis*
- Dead cattle
- Excavation
- Biological warfare
- Cutaneous route
- Aerogenic route
- Necrotic eschar
- Edema



Cutaneous Anthrax, West Bengal, India, 2007

- Cases of cutaneous anthrax, Mushidabad district, West Bengal, India, 2007.
- Dates indicate onset of skin lesion. Arrows indicate dates cattle were slaughtered.



- Black bars, cases in Sarkarpura village;
- gray bars, cases in Charbinpara village;
- white bars, deaths.

Tapas K. Ray Emerg Inf Dis 2009

Anthrax



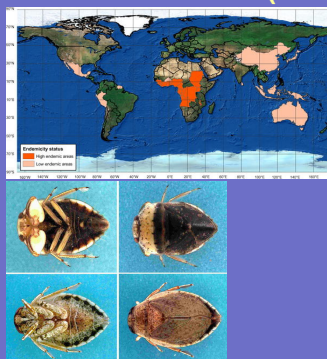
Mycobacterial ulcers

- *M. tuberculosis*
 - *Lupus vulgaris*
 - *Scrofuloderma*
- Atypical mycobacteria
 - *M. ulcerans* (Buruli ulcer)
 - *M. marinum* (Swimming pool granuloma)
- Diagnosis:
 - PCR
 - Cultivation



© Tigoulet, 2003

M. ulcerans (Buruli ulcer)



Semi-Aquatic Hemiptera positive for *M. ulcerans*





M. marinum (Swimming pool granuloma)

- Non chlorinated swimming pools
- Contact with fresh brackish and salt water fish



Diagnostic dilemmas I

- 60 y.o Surinamese women
- Since 19 yrs ulcerative and fistulating skin lesion lower leg
- Previously diagnosed as:
 - Deep mycosis - *S. Schenkii*
 - Botryomycosis- *S. aureus*
- Final diagnosis
 - Non-tuberculous mycobacterial infection - *M. marinum*



Diagnostic dilemmas II

- 45 y.o. male veterinarian
- Erythematous ulcerative plaque with crusts hand.
- Multiple systemic antibiotics and antimycotics w/o result
- Diagnosed with deep mycosis
 - 4 different species: *Aspergillus niger*, *Penicillium*, *Mucor*, and *Paraconiothyrium* species
 - Tx: voriconazol (initial improvement)
- Co-infected with:
 - Non-tuberculous mycobacterium



31 year old lady on a trip through India and China

- One night in China awake with a blister and burning pain in lower leg



Spider bite

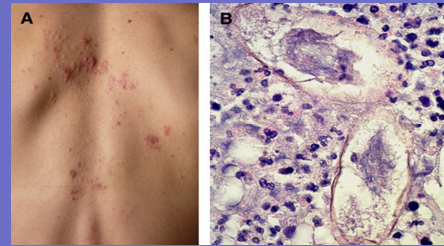
- The brown recluse or violin spider (*Loxosceles reclusa*)
 - Venom contains lipase
 - Extensive tissue destruction
 - Deep ulceration slow healing
- Steapoda (black widows)
- Tarantula (hairy spider)
 - Urticarial reaction on hairs
 - Can also bite
 - Histamine release but little tissue destruction



Case

- 28 year old female
- Had been swimming in Lake Malawi frequently
- After 5 mo. itchy papular lesions
- Abdominal cramps and flatulence

?

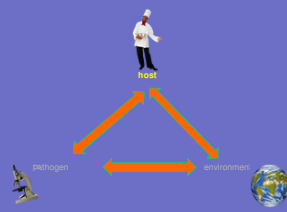


Cutaneous schistosomiasis

Van Dijk, AM J Trop Med Hyg 2010

Katayama syndrome

- Host immunological urticarial response to schistosomiasis infection



Pulmonaire urticarische afwijking op de romp in rechter afbeelding.

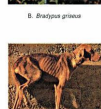
Leishmaniasis

- Protozoan infection
- Host- Mammals (inc. man)
- Vector
- Emerging infection (WHO standards)
- Clinic
 - Cutaneous (CL) ulceration, scarring
 - Mucocutaneous (MCL) cartilage destruction
 - Visceral (VL) severe internal infection (deadly)



Transmission modes

Zoonotic transmission



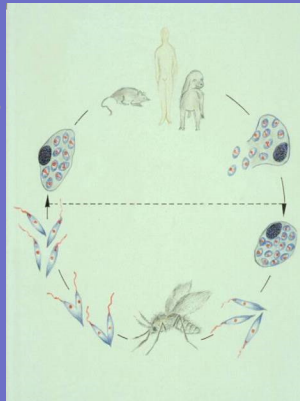
C. Vulpes vulpes D. Canis familiaris

(Antroponotic transmission)

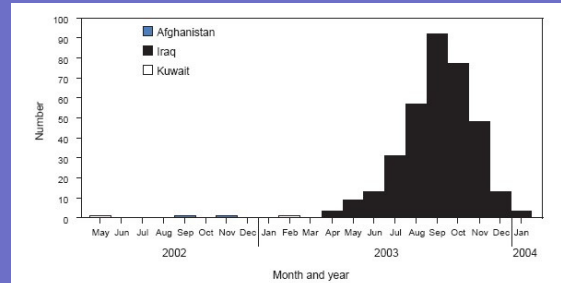


sand fly

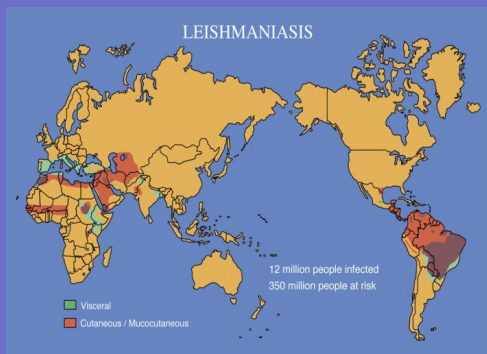
Life cycle of the *Leishmaniae*



Cutaneous leishmaniasis in U.S. military personnel 2002-2004

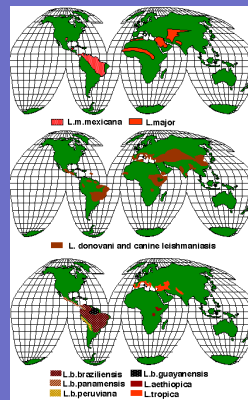


Leishmaniasis visceral and cutaneous distribution



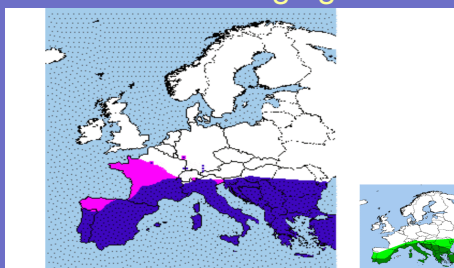
copyright: www.wehi.edu.au

Species distribution (muco)cutaneous leishmaniasis



copyright: www.microbiology.wustl.edu

Leishmaniasis emerging in Europe



Blue = distribution of visceral leishmaniasis
Red = distribution of the visceral leishmaniasis vector sandflies
Dark green = distribution of cutaneous leishmaniasis
Light green = distribution of cutaneous leishmaniasis vector sandflies
Note: Cases of VL reported from the Paris metropolitan area were probably the result of infection elsewhere.

Cutaneous leishmaniasis: Old World

- *L. major*: zoonotic (rural)
- *L. infantum*
 - * self healing, less scarring, no recidives
- *L. aethiopica*
- *L. tropica*: anthroponotic (urban)
 - * persistent, scarring, recidives



CL first stage



CL second stage



CL third stage



CL fourth stage



CL pre treatment

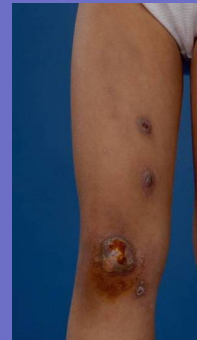


CL post treatment



cutaneous leishmaniasis

CL nodular lymphangitis



mucocutaneous leishmaniasis



Diagnostics

- Direct smear (giemsa)
- Skin biopsy
- PCR
- Cultivation



Treatment

- Local:
 - stibogluconate (Pentostam, antimony) intralesional
 - Liquid nitrogen
 - Coagulation
- Systemic
 - stibogluconate
 - pentamidine
 - miltefosine
 - itraconazole

Overview

- Skin anatomy and function
- Skin types and skin diseases
- Climate and skin
- Exanthema
- Ulcers
- Itch

Itch after visiting Curaçao



Casus

- Mrs. K, 33 years
- Flight attendant
- Since 2 months itchy papules upper leg
- After visiting Curaçao



Patient history

- Itch and burning sensations
- Esp. at night
- Skin signs disappear and reappear
- Cough
- Bloody stool
- 5 kg weight loss



Medical history

- Allergy Ni^{2+}
- Sibling allergic (atopic) eczema
- Medication prescribed by GP:
 - Prednisolone orally
 - Antihistaminics

Your diagnosis?



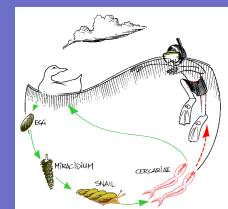
- Eczema
- IgE specific antibodies negative
- Contact allergy?



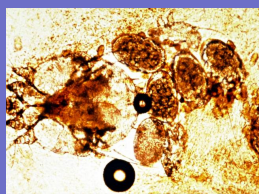
- Persistent insect bites.



- Swimmers itch?



- Scabies/schurft?



- Creeping eruption/
- Larva currens

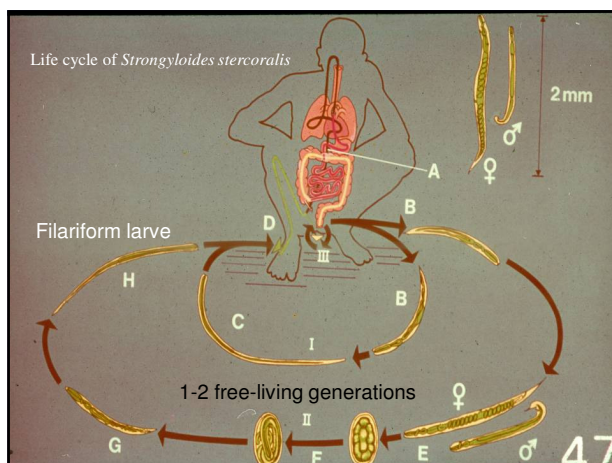


Diagnostics

- Stool examination:
 - No parasites, no eggs
- Anchylostoma serology: negative
- Strongyloides serology:
 - O.D. 161 (cut off 220),
 - One month later O.D. 251

Diagnosis

- Larva currens due to strongyloidiasis
- Therapy:
 - Ivermectin 200 microg/kg (app. 12 mg once)
 - Albendazol 400 mg dd 3 days



Prevalence

Present in most tropical and subtropical countries

Especially also South America (Surinam!) and South East Asia (Indonesia).

Also in rural areas Spain, Italia, Poland, Romania

Symptomatology

Acute phase:	itching on spot of penetration, pneumonitis (not severe) moderate to severe diarrhea
Chronic phase:	larva currens malabsorption, diarrhea Heavy infection: ulceration of small and large bowel, symptoms suggestive duodenal ulcer/ ulcerative colitis
Hyperinfection:	Fever, enteritis, ileus, shock, dyspnea, wheezing, hemoptysis, cough, meningitis

Larva currens



FIGUUR 1. Larva currens bij patiënt B (a, b): zich vrij snel verplaatsende urticariele afwijking op de rechter bil, onder 2 belichtingen opgenomen.

Strongyloid track: 5-10 cm/hour, present several days

Often no larvae in stool, serology most often positive

Birma railroad



Study in Netherlands 1990

33% (47 out of 145) ex-prisoners of war
had strongyloidiasis
45 year after leaving the tropics.

The major complication:

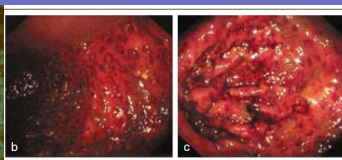
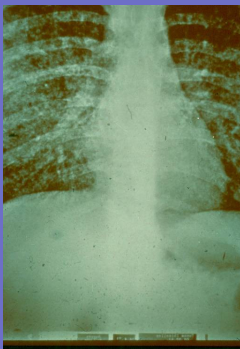
hyperinfection syndrome

Immunosuppression results in massive
proliferation of *Strongyloides stercoralis*!

First damage of gastrointestinal tract and lungs
Later on involvement of any organ: liver, heart,
pancreas, kidneys, central nervous system

Without prompt treatment: lethal condition!

hyperinfection syndrome



BREAK



Persistent insect bites

• persistent local reaction after bite, sting or contact with an "insect"

- Arthropods
 - mosquitoes
 - (sand) flies
 - scabies mites



persistent insect bites (dd?)



persistent insect bites secondary impetiginated



Persitent insect bites

1st treatment options:

- local corticosteroid (+ hydrocolloid occlusion)
- antihistaminics

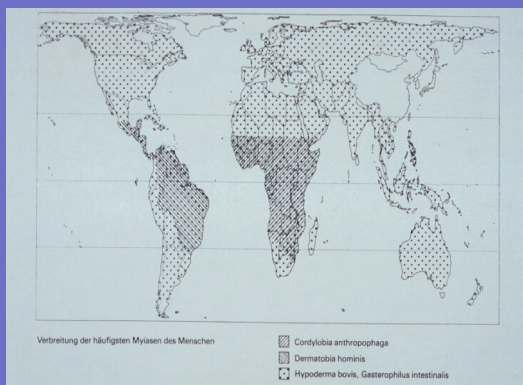
2nd options:

- oral prednisolone
- thalidomide
- ivermectine

Cutaneous myiasis

- wound myiasis
- subcutaneous myiasis
- dermal myiasis

Wound myiasis



Subcutaneous myiasis

- Skin penetrating larvae
- Nodi with central opening
- Maximal duration ca 6-12 weeks
- Unprotected skin
- Tx: suffocate with vaseline?
- Biopsy and GENTLE extirpation



Subcutaneous myiasis

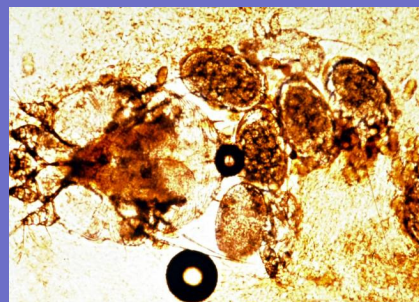


Portuguese man o' war jellyfish (Physalia)

- Potentially deadly
 - can cause digital gangrene
- Misconceptions on treatment:
 - Rinse with vinegar or,
 - Rinse with hot water (45°C) or,
 - Hot compress 45 minutes
 - Then, remove remaining tentacles with tweezers)



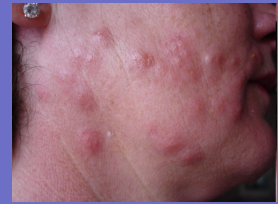
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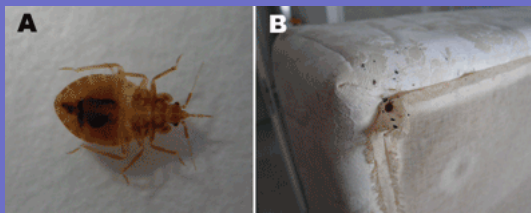
scabies



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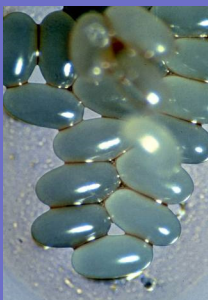
Bedbug dermatitis



Delaunay P et al. Emerg Infect Dis 2009



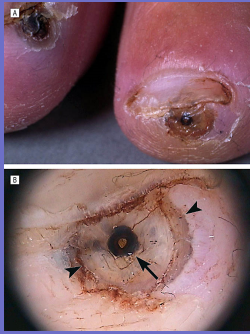
Tunga penetrans



Tungiasis

- Sandflea (*Tunga penetrans*)
- Endemic in tropical regions
- Esp. sandy soil
- Cows, pigs
- Blood meal
- Female penetrates foot soles lays ca. 100 eggs.

A, Subungual tungiasis

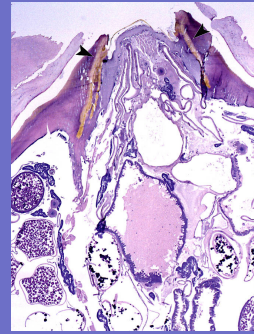


Bauer, J. et al. Arch Dermatol 2004;140:761-763.

ARCHIVES OF
DERMATOLOGY

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Histopathologic image of Tunga penetrans showing the partially brown pigmented chitin of the exoskeleton (arrowheads), which corresponds to the pigmented ring seen under dermoscopy (hematoxylin-eosin, original magnification x100)



Bauer, J. et al. Arch Dermatol 2004;140:761-763.

ARCHIVES OF
DERMATOLOGY

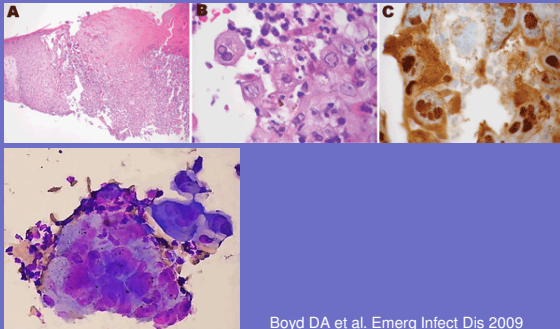
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eczema herpeticum mimicking small pox



Boyd DA et al. Emerg Infect Dis 2009

Small pox



dd?

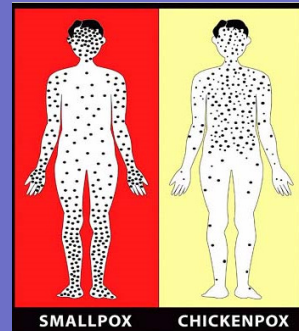


small pox chicken pox



(varicella zoster primary infection)

centrifugal centripetal

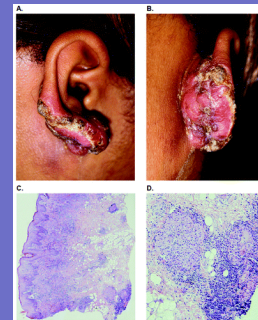


For your ears only



Differential diagnosis?

LUPUS VULGARIS



Aguillar, Clin Inf Dis 2010

Chiclero's ulcer



- Mucocutaneous leishmaniasis
- Typical for Middle American region
- Mainly *L. mexicana* (*L. panamensis*)

Tropical Disease Unit, Toronto General Hospital

Lepromatous leprosy



Lyme borreliosis

Erythema chronicum Migrans



Lyme disease

- Erythema chronicum migrans
- Acrodermatitis chronica atrophicans



Ant-induced alopecia



Namazi, M. R. et al. Arch Dermatol 2008;144:1526-1527.

Cut hairs and ants brought by the patient



Namazi, M. R. et al. Arch Dermatol 2008;144:1526-1527.

